

03 Main Analyses

The ‘Immigration Issue’ Misnomer in Human Capital Research:
Intersectional Inequalities in Adult Education and Literacy in Upper- and
Middle-Income Countries

Author Omitted for Blind Peer Review

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Setup

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)
```

```
packages <- c('tidyverse',  
              'readr',  
              'knitr',  
              'here',  
              'ggplot2',  
              'ggrepel',  
              'ragg',  
              'kableExtra',  
              'countrycode',  
              'readxl',  
              'jtools',  
              'lme4',  
              'lmerTest',  
              'broom.mixed',  
              'viridis',  
              'survey',  
              'ggpubr',  
              'purrr')
```

```
pacman::p_load(packages, character.only = T)
```

Data

```
#df <- read_rds(here("Data", "df.RDS"))  
df2 <- read_rds(here("Data", "df2.RDS"))
```

Regressions

Table A2. OLS

```
df_imm <- subset(df2, FORBORN == 1 & !is.na(NFE12) & !is.na(children) &  
  !is.na(FORLANG) & !is.na(ed3) & ed_currently != 1 & !is.na(female))  
  
df_nat <- subset(df2, FORBORN == 0 & !is.na(NFE12) & !is.na(children) &  
  !is.na(FORLANG) & !is.na(ed3) & ed_currently != 1 & !is.na(female))  
  
df_clean <- subset(df2, !is.na(NFE12) & !is.na(children) & !is.na(FORLANG) &  
  !is.na(ed3) & ed_currently != 1 & !is.na(female))  
  
M1.1 <- lm(skill_reading ~ NFE12 + female + children + FORLANG + factor(ed3)  
  + factor(iso3c), data = df_imm)  
  
M1.2 <- lm(skill_reading ~ NFE12 + female + children + FORLANG + factor(ed3)  
  + factor(iso3c), data = df_nat)  
  
M1.3 <- lm(skill_reading ~ NFE12 + female + children + FORLANG + FORBORN +
```

```

factor(ed3) + factor(iso3c), data = df_clean)

M1.4 <- lm(skill_reading ~ NFE12 + female + children + FORLANG +
FORBORN*NFE12 + factor(ed3) + factor(iso3c), data = df_clean)

export_summs(
  M1.1, M1.2, M1.3, M1.4,
  model.names = c("M1.1", "M1.2", "M1.3", "M1.4"),
  coefs = c(
    "(Intercept)",
    "NFE12",
    "female",
    "children",
    "FORLANG",
    "FORBORN",
    "factor(ed3)2",
    "factor(ed3)3",
    "NFE12:FORBORN"
  ),
  statistics = c("N" = "nobs", "r2" = "r.squared", "AIC" = "AIC", "BIC" =
"BIC"),
  to.file = "docx",
  file.name = here("Results", "TA2.docx")
)

```

	M1.1	M1.2	M1.3	M1.4
(Intercept)	201.12 *** (2.32)	225.46 *** (0.87)	224.01 *** (0.81)	224.27 *** (0.81)
NFE12	20.97 *** (0.84)	15.48 *** (0.28)	16.29 *** (0.27)	15.81 *** (0.29)
female	-0.14 (0.78)	-0.66 * (0.26)	-0.60 * (0.25)	-0.58 * (0.25)
children	-10.10 *** (0.86)	-7.81 *** (0.28)	-8.13 *** (0.27)	-8.12 *** (0.27)
FORLANG	-17.58 *** (0.88)	-12.63 *** (0.75)	-16.82 *** (0.52)	-16.79 *** (0.52)
FORBORN			-17.52 *** (0.45)	-19.04 *** (0.55)
factor(ed3)2	30.27 *** (1.19)	30.45 *** (0.43)	30.75 *** (0.41)	30.72 *** (0.41)
factor(ed3)3	63.02 *** (1.21)	61.43 *** (0.44)	61.73 *** (0.42)	61.71 *** (0.42)
NFE12:FORBORN				3.61 *** (0.74)
N	15914	106727	122641	122641
r2	0.31	0.37	0.37	0.37
AIC	168732.95	1099077.30	1269391.94	1269370.44
BIC	169001.58	1099412.53	1269741.76	1269729.97

*** p < 0.001; ** p < 0.01; * p < 0.05.

Sample Bias Test

Here I correlate the residuals with the weighting variable. If this is close to zero I can look at the sample as a representative treatment for ALE and literacy.

```
df_imm$resid_imm_1 <- residuals(M1.1)
df_nat$resid_nat_1 <- residuals(M1.2)
df_clean$resid_clean <- residuals(M1.3)

testcor1 <- cor(df_imm$aweight, df_imm$resid_imm_1)
testcor2 <- cor(df_nat$aweight, df_nat$resid_nat_1)
testcor3 <- cor(df_clean$aweight, df_clean$resid_clean)

testp1 <- cor.test(df_imm$aweight, df_imm$resid_imm_1)$p.value
testp2 <- cor.test(df_nat$aweight, df_nat$resid_nat_1)$p.value
testp3 <- cor.test(df_clean$aweight, df_clean$resid_clean)$p.value
```

Correlations are 0.014, 0.012, & 0.009. With p-values 0.077, 10^{-4} , & 0.0026. This suggests that there is no need to weight the sample. We are essentially controlling for the variables that lead to selection.

Table A3. GLM/Multilevel

Recodes

```
df2_sub <- df2 %>%
  mutate(
    ed3_secondary = as.integer(ed3 == 2),
    ed3_tertiary = as.integer(ed3 == 3)
  ) %>%
  subset(ed_currently != 1)

# within/between decomposition (weighted country means)
df2_sub <- df2_sub %>%
  group_by(iso3c) %>%
  mutate(
    male = ifelse(female == 1, 0, 1),
    NFE12_between = sum(SPFWT0 * NFE12, na.rm = TRUE) /
sum(SPFWT0[!is.na(NFE12)], na.rm = TRUE),
    female_between = sum(SPFWT0 * female, na.rm = TRUE) /
sum(SPFWT0[!is.na(female)], na.rm = TRUE),
    male_between = sum(SPFWT0 * male, na.rm = T) / sum(SPFWT0[!is.na(male)],
na.rm = TRUE),
    children_between = sum(SPFWT0 * children, na.rm = TRUE) /
sum(SPFWT0[!is.na(children)], na.rm = TRUE),
    ed3_secondary_between = sum(SPFWT0 * ed3_secondary, na.rm = TRUE) /
sum(SPFWT0[!is.na(ed3_secondary)], na.rm = TRUE),
    ed3_tertiary_between = sum(SPFWT0 * ed3_tertiary, na.rm = TRUE) /
sum(SPFWT0[!is.na(ed3_tertiary)], na.rm = TRUE),

    forbourn_between = sum(SPFWT0 * FORBORN, na.rm = TRUE) /
sum(SPFWT0[!is.na(FORBORN)], na.rm = TRUE),
    forlang_between = sum(SPFWT0 * FORLANG, na.rm = TRUE) /
sum(SPFWT0[!is.na(FORLANG)], na.rm = TRUE),

    NFE12_within = NFE12 - NFE12_between,
    children_within = children - children_between,
    female_within = female - female_between,
    ed3_secondary_within = ed3_secondary - ed3_secondary_between,
    ed3_tertiary_within = ed3_tertiary - ed3_tertiary_between,
    forbourn_within = FORBORN - forbourn_between,
    forlang_within = FORLANG - forlang_between
  ) %>%
  ungroup()

# the within effects need to be further centered by group

df2_imm <- df2_sub %>%
  subset(FORBORN == 1) %>%
  group_by(iso3c) %>%
  mutate(NFE12_within = NFE12_within - mean(NFE12_within, na.rm = TRUE),
    female_within = female_within - mean(female_within, na.rm = T),
```

```

      children_within = children_within - mean(children_within, na.rm =
TRUE),
      ed3_secondary_within = ed3_secondary_within -
mean(ed3_secondary_within, na.rm = TRUE),
      ed3_tertiary_within = ed3_tertiary_within -
mean(ed3_tertiary_within, na.rm = TRUE) ) %>%
    ungroup()

df2_nat <- df2_sub %>%
  subset(FORBORN_not == 1) %>%
  group_by(iso3c) %>%
  mutate(NFE12_within = NFE12_within - mean(NFE12_within, na.rm = TRUE),
    children_within = children_within - mean(children_within, na.rm =
TRUE),
    female_within = female_within - mean(female_within, na.rm = T),
    ed3_secondary_within = ed3_secondary_within -
mean(ed3_secondary_within, na.rm = TRUE),
    ed3_tertiary_within = ed3_tertiary_within -
mean(ed3_tertiary_within, na.rm = TRUE) ) %>% ungroup()

between_vars <- c(
  "NFE12_between", "female_between", "children_between",
  "ed3_secondary_between", "ed3_tertiary_between",
  "forborn_between", "forlang_between"
)

# build country-level x-standardized-scored between
country_between_scaled <- function(dat, between_vars, suffix = "_z") {

  # get one row per country (so no implicit weighting by country sample size)
  country_vals <- dat %>%
    distinct(iso3c, across(all_of(between_vars)))

  # z-score across countries
  country_z <- country_vals %>%
    mutate(
      across(all_of(between_vars), ~{
        m <- mean(.x, na.rm = TRUE)
        s <- sd(.x, na.rm = TRUE)
        if (is.na(s) || s == 0) NA_real_ else as.numeric((.x - m) / s)
      }, .names = paste0("{.col}", suffix))
    ) %>%
    select(iso3c, ends_with(suffix))

  country_z
}

df2_sub <- df2_sub %>%
  left_join(country_between_scaled(df2_sub, between_vars), by = "iso3c")

```

```
df2_imm <- df2_imm %>%
  left_join(country_between_scaled(df2_imm, between_vars), by = "iso3c")

df2_nat <- df2_nat %>%
  left_join(country_between_scaled(df2_nat, between_vars), by = "iso3c")
```

Models

I first look at immigrants and native-born separately. Then combine them to determine if there is an ‘immigrant penalty’ after controlling for participation in further education and other variables.

```
M2.1 <- lmer(
  skill_reading ~
    NFE12_within + NFE12_between_z +
    female_within + female_between_z +
    children_within + children_between_z +
    forlang_within + forlang_between_z +
    ed3_secondary_within + ed3_secondary_between_z +
    ed3_tertiary_within + ed3_tertiary_between_z +
    (1 | iso3c),
  data = df2_imm
)
```

```
ranef_list <- ranef(M2.1)
```

```
# Get country-level grand intercept deviations
iso3c_intercepts <- ranef_list$iso3c
```

```
M2.2 <- lmer(
  skill_reading ~
    NFE12_within + NFE12_between_z +
    female_within + female_between_z +
    children_within + children_between_z +
    forlang_within + forlang_between_z +
    ed3_secondary_within + ed3_secondary_between_z +
    ed3_tertiary_within + ed3_tertiary_between_z +
    (1 | iso3c),
  data = df2_nat
)
```

```
M2.3a <- lmer(
  skill_reading ~
    NFE12_within + NFE12_between_z +
    children_within + children_between_z +
    female_within + female_between_z +
    forlang_within + forlang_between_z +
```

```

    forborn_within + forborn_between_z +
    ed3_secondary_within + ed3_secondary_between_z +
    ed3_tertiary_within + ed3_tertiary_between_z +
    (1 | iso3c),
    data = subset(df2_sub)
)

```

```

M2.3b <- lmer(
  skill_reading ~
    NFE12_within + NFE12_between_z +
    children_within + children_between_z +
    female_within + female_between_z +
    forlang_within + forlang_between_z +
    forborn_within + forborn_between_z +
    ed3_secondary_within + ed3_secondary_between_z +
    ed3_tertiary_within + ed3_tertiary_between_z +
    gdppc_k +
    (1 | iso3c),
    data = subset(df2_sub)
)

```

```

M2.4 <- lmer(
  skill_reading ~
    NFE12_within + NFE12_between_z +
    children_within + children_between_z +
    female_within + female_between_z +
    forlang_within + forlang_between_z +
    forborn_within + forborn_between_z +
    forborn_within*NFE12_within +
    ed3_secondary_within + ed3_secondary_between_z +
    ed3_tertiary_within + ed3_tertiary_between_z +
    (1 | iso3c),
    data = subset(df2_sub)
)

```

Results

```

export_summs(
  M2.1, M2.2, M2.3a, M2.3b, M2.4,
  #model.names = c("Foreign-born", "Native-born", "Pooled", "Pooled with ALE
interaction"),
  to.file = "docx",
  file.name = here("Results", "TA3.docx"
))

```

	Model 1	Model 2	Model 3	Model 4	Model 5
(Intercept)	245.49 *** (2.72)	263.14 *** (2.65)	258.48 *** (2.36)	256.56 *** (16.60)	258.50 *** (2.36)
NFE12_within	20.97 *** (0.84)	15.48 *** (0.28)	16.29 *** (0.27)	16.29 *** (0.27)	16.42 *** (0.27)
NFE12_between_z	15.62 *** (3.69)	13.33 ** (3.63)	11.73 ** (3.65)	11.64 ** (3.82)	11.73 ** (3.65)
female_within	-0.14 (0.78)	-0.66 * (0.26)	-0.60 * (0.25)	-0.60 * (0.25)	-0.58 * (0.25)
female_between_z	-0.79 (3.27)	-7.05 * (3.26)	-6.46 * (2.91)	-6.38 * (3.05)	-6.47 * (2.91)
children_within	-10.10 *** (0.86)	-7.81 *** (0.28)	-8.13 *** (0.27)	-8.13 *** (0.27)	-8.11 *** (0.27)
children_between_z	-8.37 * (3.51)	-7.51 * (3.45)	-7.37 * (3.16)	-7.24 * (3.44)	-7.37 * (3.16)
forlang_within	-17.61 *** (0.88)	-12.63 *** (0.75)	-16.82 *** (0.52)	-16.82 *** (0.52)	-16.75 *** (0.52)
forlang_between_z	-3.46 (3.47)	-1.27 (3.43)	-1.47 (3.86)	-1.63 (4.19)	-1.46 (3.86)
ed3_secondary_within	30.27 *** (1.19)	30.45 *** (0.43)	30.75 *** (0.41)	30.75 *** (0.41)	30.74 *** (0.41)
ed3_secondary_between_z	7.11 (3.56)	1.96 (3.48)	3.02 (3.52)	2.92 (3.71)	3.00 (3.53)
ed3_tertiary_within	63.02 *** (1.21)	61.43 *** (0.44)	61.73 *** (0.42)	61.73 *** (0.42)	61.74 *** (0.42)
ed3_tertiary_between_z	-2.09 (3.83)	0.49 (3.82)	2.12 (3.86)	1.96 (4.20)	2.10 (3.86)
forborn_within			-17.52 *** (0.45)	-17.52 *** (0.45)	-17.49 *** (0.45)
forborn_between_z			-2.06 (4.32)	-2.24 (4.68)	-2.06 (4.32)
gdppc_k				0.05 (0.42)	
NFE12_within:forborn_within					5.41 *** (0.79)

N	15914	106727	122641	122641	122641
N (iso3c)	28	28	28	28	28
AIC	168770.48	1099199.65	1269509.37	1269511.28	1269462.90
BIC	168885.61	1099343.32	1269674.56	1269686.18	1269637.81
R2 (fixed)	0.27	0.32	0.33	0.33	0.33
R2 (total)	0.33	0.39	0.38	0.38	0.38

*** p < 0.001; ** p < 0.01; * p < 0.05.

GDP?

The correlation is significant at $p < 0.1$ without the US. The US is kind of a disaster in this figure.

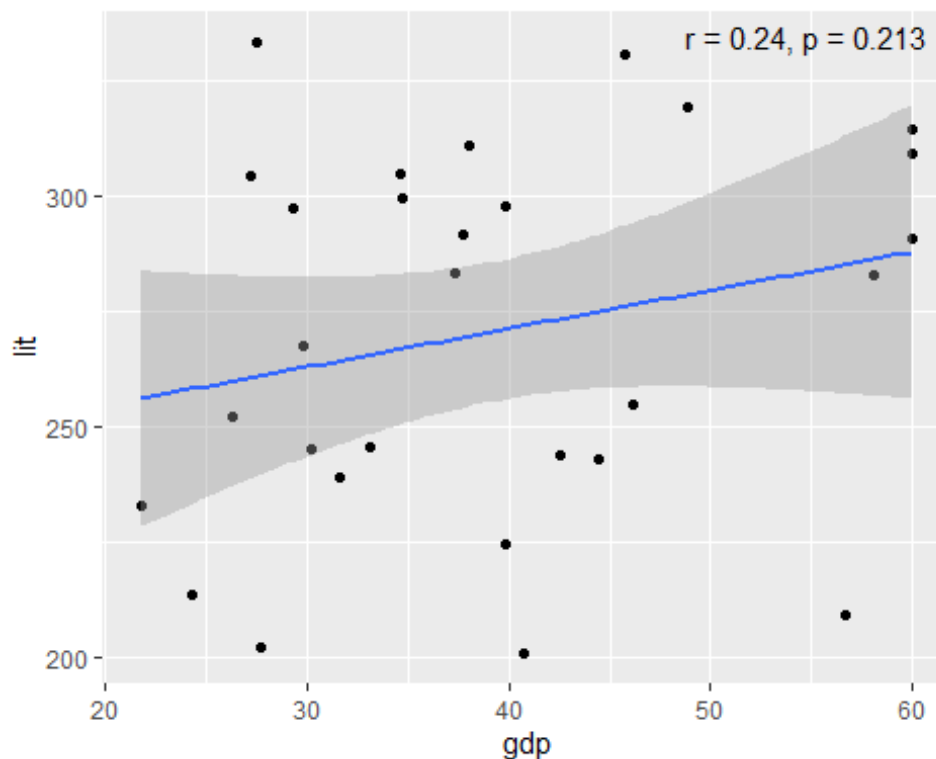
```
test <- df2_sub %>%
  group_by(iso3c) %>%
  summarise(lit = first(skill_reading), gdp = first(gdppc_k))

test_nousa <- subset(test, iso3c != "USA")

ct <- cor.test(test$gdp, test$lit)
ctusa <- cor.test(test_nousa$gdp, test_nousa$lit)

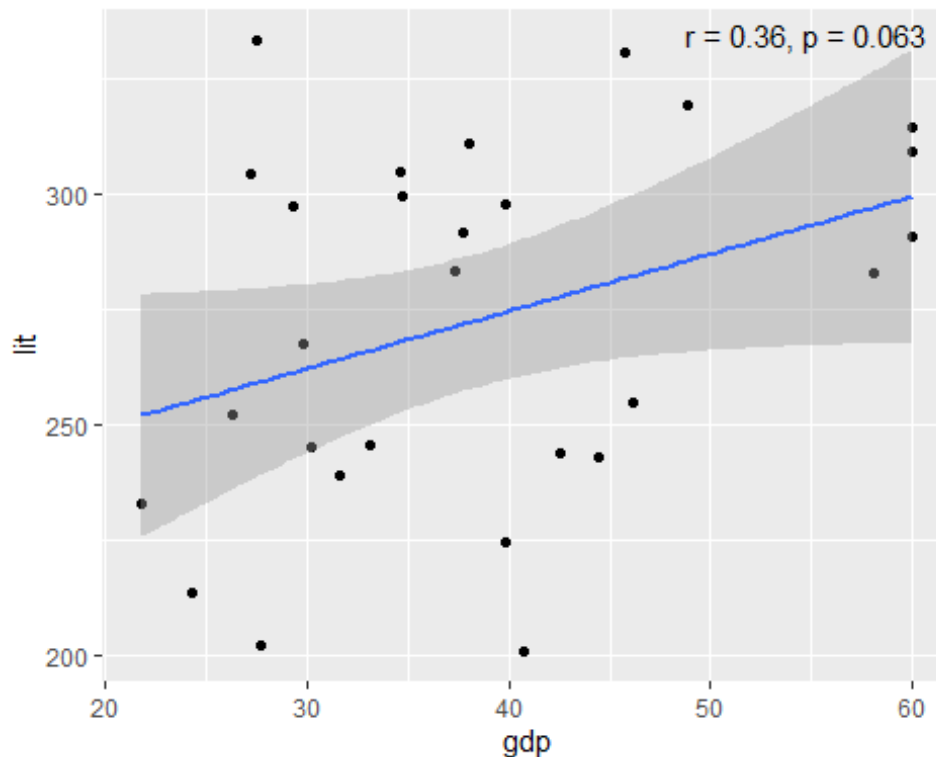
lab <- sprintf("r = %.2f, p = %.3f", ct$estimate, ct$p.value)
labusa <- sprintf("r = %.2f, p = %.3f", ctusa$estimate, ctusa$p.value)

ggplot(test, aes(gdp, lit)) +
  geom_point() +
  geom_smooth(method = "lm") +
  annotate("text",
    x = Inf, y = Inf,
    label = lab,
    hjust = 1.1, vjust = 1.5)
```



```
# without the US
ggplot(test_nousa, aes(gdp, lit)) +
```

```
geom_point() +
geom_smooth(method = "lm") +
annotate("text",
  x = Inf, y = Inf,
  label = labusa,
  hjust = 1.1, vjust = 1.5)
```



ATEs

Table A4. Interactive GLM

Here we interact all intersectional variables of interest with ALE (NFE12) to see the unique effects if any.

The correlation between foreign-born and foreign-language is not so high as to warrant dropping one.

```
select(df2, FORBORN, FORLANG) %>% cor(., use = "pairwise.complete.obs")
FORBORN FORLANG FORBORN 1.0000000 0.5645113 FORLANG 0.5645113 1.0000000
```

```
m4 <- lmer(
  skill_reading ~
    NFE12_within + NFE12_between_z +
    female_within + female_between_z +
    children_within + children_between_z +
    forlang_within + forlang_between_z +
    forborn_within + forborn_between_z +
```

```

# interaction with all sub-interactions
female_within*forborn_within*children_within*forlang_within*NFE12_within
+
ed3_secondary_within + ed3_secondary_between_z +
ed3_tertiary_within + ed3_tertiary_between_z +
gdppc_k_log +
(1 | iso3c),
data = subset(df2_sub)
)

export_summs(
  m4,
  coefs = c(
    "(Intercept)" = "(Intercept)",
    "ALE (within)" = "NFE12_within",
    "ALE (between)" = "NFE12_between_z",
    "Female (within)" = "female_within",
    "Female (between)" = "female_between_z",
    "Children (within)" = "children_within",
    "Children (between)" = "children_between_z",
    "Non-native speaker (within)" = "forlang_within",
    "Non-native speaker (between)" = "forlang_between_z",
    "Foreign-born (within)" = "forborn_within",
    "Foreign-born (between)" = "forborn_between_z",
    "Secondary education (within)" = "ed3_secondary_within",
    "Secondary education (between)" = "ed3_secondary_between_z",
    "Tertiary education (within)" = "ed3_tertiary_within",
    "Tertiary education (between)" = "ed3_tertiary_between_z",
    "GDP per capita (log k)" = "gdppc_k_log",
    "Female x Foreign-born (within)" = "female_within:forborn_within",
    "Female x Children (within)" = "female_within:children_within",
    "Children x Foreign-born (within)" = "children_within:forborn_within",
    "Female x Foreign-language (within)" = "female_within:forlang_within",
    "Foreign-language x Foreign-born (within)" =
"forlang_within:forborn_within",
    "Children x Foreign-language (within)" =
"children_within:forlang_within",
    "ALE x Female (within)" = "NFE12_within:female_within",
    "ALE x Foreign-born (within)" = "NFE12_within:forborn_within",
    "ALE x Children (within)" = "NFE12_within:children_within",
    "ALE x Foreign-language (within)" = "NFE12_within:forlang_within",
    "Female x Children x Foreign-born" =
"female_within:children_within:forborn_within",
    "Female x Foreign-language x Foreign-born (within)" =
"female_within:forlang_within:forborn_within",
    "Female x Children x Foreign-language (within)" =
"female_within:children_within:forlang_within",
    "Children x Foreign-language x Foreign-born (within)" =
"children_within:forlang_within:forborn_within",

```

```

    "ALE x Female x Foreign-born (within)" =
"NFE12_within:female_within:forborn_within",
    "ALE x Female x Children (within)" =
"NFE12_within:female_within:children_within",
    "ALE x Children x Foreign-born (within)" =
"NFE12_within:children_within:forborn_within",
    "ALE x Female x Foreign-language (within)" =
"NFE12_within:female_within:forlang_within",
    "ALE x Foreign-language x Foreign-born (within)" =
"NFE12_within:forlang_within:forborn_within",
    "ALE x Children x Foreign-language (within)" =
"NFE12_within:children_within:forlang_within",
    "Female x Children x Foreign-language x Foreign-born (within)" =
"female_within:children_within:forlang_within:forborn_within",
    "ALE x Female x Children x Foreign-born (within)" =
"NFE12_within:female_within:children_within:forborn_within",
    "ALE x Female x Foreign-language x Foreign-born (within)" =
"NFE12_within:female_within:forlang_within:forborn_within",
    "ALE x Female x Children x Foreign-language (within)" =
"NFE12_within:female_within:children_within:forlang_within",
    "ALE x Children x Foreign-language x Foreign-born" =
"NFE12_within:children_within:forlang_within:forborn_within",
    "ALE x Female x Children x Foreign-language x Foreign-born" =
"NFE12_within:female_within:children_within:forlang_within:forborn_within"
),
statistics = "all",
error_pos = "right",
error_format = "({std.error})", # Formats the S.E. column (e.g., with
parentheses)
to.file = "docx",
file.name = here("Results", "TA4.docx")
)

```

	Model 1	
(Intercept)	217.01 **	(61.47)
ALE (within)	16.06 ***	(0.30)
ALE (between)	11.17 **	(3.80)
Female (within)	-0.69 *	(0.27)
Female (between)	-6.04	(3.04)
Children (within)	-7.99 ***	(0.30)
Children (between)	-6.39	(3.53)
Non-native speaker (within)	-15.84 ***	(0.62)
Non-native speaker (between)	-2.36	(4.14)
Foreign-born (within)	-16.94 ***	(0.48)
Foreign-born (between)	-2.84	(4.59)
Secondary education (within)	30.65 ***	(0.41)
Secondary education (between)	2.53	(3.66)
Tertiary education (within)	61.71 ***	(0.42)
Tertiary education (between)	1.14	(4.17)
GDP per capita (log k)	11.52	(16.94)
Female x Foreign-born (within)	-0.85	(0.94)

Female x Children (within)	-1.90 **	(0.59)
Children x Foreign-born (within)	-1.81	(1.03)
Female x Foreign-language (within)	2.90 *	(1.21)
Foreign-language x Foreign-born (within)	-3.39 **	(1.06)
Children x Foreign-language (within)	-0.19	(1.28)
ALE x Female (within)	-3.44 ***	(0.58)
ALE x Foreign-born (within)	2.82 **	(0.97)
ALE x Children (within)	-2.82 ***	(0.62)
ALE x Foreign-language (within)	0.50	(1.24)
Female x Children x Foreign-born	0.04	(2.05)
Female x Foreign-language x Foreign-born (within)	2.44	(1.94)
Female x Children x Foreign-language (within)	3.23	(2.55)
Children x Foreign-language x Foreign-born (within)	-3.47	(2.06)
ALE x Female x Foreign-born (within)	0.74	(1.95)
ALE x Female x Children (within)	-1.06	(1.24)
ALE x Children x Foreign-born (within)	2.57	(2.11)
ALE x Female x Foreign-language (within)	0.90	(2.49)
ALE x Foreign-language x Foreign-born (within)	8.06 ***	(2.03)
ALE x Children x Foreign-language (within)	-2.87	(2.59)
Female x Children x Foreign-language x Foreign-born (within)	5.52	(4.12)
ALE x Female x Children x Foreign-born (within)	0.18	(4.22)
ALE x Female x Foreign-language x Foreign-born (within)	-0.80	(4.06)
ALE x Female x Children x Foreign-language (within)	-7.84	(5.19)
ALE x Children x Foreign-language x Foreign-born	8.62 *	(4.28)
ALE x Female x Children x Foreign-language x Foreign-born	16.38	(8.56)
nobs	122641	
nobs.1	122641.00	
sigma	42.75	
logLik	-634585.81	
AIC	1269259.63	
BIC	1269687.18	
REMLcrit	1269171.63	
df.residual	122597.00	
p.value		
r.squared	0.38	
r.squared.fixed	0.33	
group.nobs.iso3c	28.00	

*** p < 0.001; ** p < 0.01; * p < 0.05.

`include_graphics(here("Results", "TA4.docx"))`

Predictions

```
df2r <- subset(df2_sub, !is.na(female) & !is.na(NFE12) &
!is.na(children_within) & !is.na(forborn_within) & !is.na(forlang_within) &
!is.na(ed3) & !is.na(gdppc) & !is.na(skill_reading))
```

Calculate grand means. This is necessary because of the centering. The between effect would otherwise create artifact effects because there are more or less immigrants (or whatever group) in a country

```

gm_female <- mean(df2r$female_between, na.rm = TRUE)
gm_forborn <- mean(df2r$forborn_between, na.rm = TRUE)
gm_children <- mean(df2r$children_between, na.rm = TRUE)
gm_forlang <- mean(df2r$forlang_between, na.rm = TRUE)
gm_nfe12 <- mean(df2r$NFE12_between, na.rm = TRUE)

#Type 1: Foreign-born parent, non-native speaker
df_type10 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
1 - gm_forborn, children_within = 1 - gm_children, forlang_within = 1 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type11 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
1 - gm_forborn, children_within = 1 - gm_children, forlang_within = 1 -
gm_forlang, NFE12_within = 1 - gm_nfe12)
df_type10m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
1 - gm_forborn, children_within = 1 - gm_children, forlang_within = 1 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type11m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
1 - gm_forborn, children_within = 1 - gm_children, forlang_within = 1 -
gm_forlang, NFE12_within = 1 - gm_nfe12)

#Type 2: Foreign-born no children, non-native speaker
df_type20 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
1 - gm_forborn, children_within = 0 - gm_children, forlang_within = 1 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type21 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
1 - gm_forborn, children_within = 0 - gm_children, forlang_within = 1 -
gm_forlang, NFE12_within = 1 - gm_nfe12)
df_type20m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
1 - gm_forborn, children_within = 0 - gm_children, forlang_within = 1 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type21m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
1 - gm_forborn, children_within = 0 - gm_children, forlang_within = 1 -
gm_forlang, NFE12_within = 1 - gm_nfe12)

#Type 3: Foreign-born parent, native speaker
df_type30 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
1 - gm_forborn, children_within = 1 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type31 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
1 - gm_forborn, children_within = 1 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 1 - gm_nfe12)
df_type30m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
1 - gm_forborn, children_within = 1 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type31m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
1 - gm_forborn, children_within = 1 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 1 - gm_nfe12)

#Type 4: Foreign-born no children, native speaker
df_type40 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =

```

```

1 - gm_forborn, children_within = 0 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type41 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
1 - gm_forborn, children_within = 0 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 1 - gm_nfe12)
df_type40m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
1 - gm_forborn, children_within = 0 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type41m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
1 - gm_forborn, children_within = 0 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 1 - gm_nfe12)

```

#Type 5: Native-born parent, native speaker

```

df_type50 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
0 - gm_forborn, children_within = 1 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type51 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
0 - gm_forborn, children_within = 1 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 1 - gm_nfe12)
df_type50m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
0 - gm_forborn, children_within = 1 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type51m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
0 - gm_forborn, children_within = 1 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 1 - gm_nfe12)

```

#Type 6: Native-born no children, native speaker

```

df_type60 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
0 - gm_forborn, children_within = 0 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type61 <- df2r %>% mutate(female_within = 1 - gm_female, forborn_within =
0 - gm_forborn, children_within = 0 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 1 - gm_nfe12)
df_type60m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
0 - gm_forborn, children_within = 0 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 0 - gm_nfe12)
df_type61m <- df2r %>% mutate(female_within = 0 - gm_female, forborn_within =
0 - gm_forborn, children_within = 0 - gm_children, forlang_within = 0 -
gm_forlang, NFE12_within = 1 - gm_nfe12)

```

make predictions

```

pred10 <- predict(m4, newdata = df_type10, re.form = NA)
pred11 <- predict(m4, newdata = df_type11, re.form = NA)
pred10m <- predict(m4, newdata = df_type10m, re.form = NA)
pred11m <- predict(m4, newdata = df_type11m, re.form = NA)
pred20 <- predict(m4, newdata = df_type20, re.form = NA)
pred21 <- predict(m4, newdata = df_type21, re.form = NA)
pred20m <- predict(m4, newdata = df_type20m, re.form = NA)
pred21m <- predict(m4, newdata = df_type21m, re.form = NA)

```

```

pred30 <- predict(m4, newdata = df_type30, re.form = NA)
pred31 <- predict(m4, newdata = df_type31, re.form = NA)
pred30m <- predict(m4, newdata = df_type30m, re.form = NA)
pred31m <- predict(m4, newdata = df_type31m, re.form = NA)
pred40 <- predict(m4, newdata = df_type40, re.form = NA)
pred41 <- predict(m4, newdata = df_type41, re.form = NA)
pred40m <- predict(m4, newdata = df_type40m, re.form = NA)
pred41m <- predict(m4, newdata = df_type41m, re.form = NA)
pred50 <- predict(m4, newdata = df_type50, re.form = NA)
pred51 <- predict(m4, newdata = df_type51, re.form = NA)
pred50m <- predict(m4, newdata = df_type50m, re.form = NA)
pred51m <- predict(m4, newdata = df_type51m, re.form = NA)
pred60 <- predict(m4, newdata = df_type60, re.form = NA)
pred61 <- predict(m4, newdata = df_type61, re.form = NA)
pred60m <- predict(m4, newdata = df_type60m, re.form = NA)
pred61m <- predict(m4, newdata = df_type61m, re.form = NA)

```

Calculate average marginal effect

```

diff1 <- pred10 - pred11
diff2 <- pred20 - pred21
diff3 <- pred30 - pred31
diff4 <- pred40 - pred41
diff5 <- pred50 - pred51
diff6 <- pred60 - pred61

```

```

diff1m <- pred10m - pred11m
diff2m <- pred20m - pred21m
diff3m <- pred30m - pred31m
diff4m <- pred40m - pred41m
diff5m <- pred50m - pred51m
diff6m <- pred60m - pred61m

```

```

AME1 <- mean(diff1)
ttest1 <- t.test(pred11, pred10, paired = FALSE)
AME2 <- mean(diff2)
ttest2 <- t.test(pred21, pred20, paired = FALSE)
AME3 <- mean(diff3)
ttest3 <- t.test(pred31, pred30, paired = FALSE)
AME4 <- mean(diff4)
ttest4 <- t.test(pred41, pred40, paired = FALSE)
AME5 <- mean(diff5)
ttest5 <- t.test(pred51, pred50, paired = FALSE)
AME6 <- mean(diff6)
ttest6 <- t.test(pred61, pred60, paired = FALSE)

```

```

AME1m <- mean(diff1m)
ttest1m <- t.test(pred11m, pred10m, paired = FALSE)
AME2m <- mean(diff2m)
ttest2m <- t.test(pred21m, pred20m, paired = FALSE)
AME3m <- mean(diff3m)
ttest3m <- t.test(pred31m, pred30m, paired = FALSE)
AME4m <- mean(diff4m)
ttest4m <- t.test(pred41m, pred40m, paired = FALSE)
AME5m <- mean(diff5m)
ttest5m <- t.test(pred51m, pred50m, paired = FALSE)
AME6m <- mean(diff6m)
ttest6m <- t.test(pred61m, pred60m, paired = FALSE)

```

Plot

Combine values

```

ame_df <- data.frame(
  type = c(
    "1f. Foreign-born mother\nnon-native speaker",
    "2f. Foreign-born no children female\nnon-native speaker",
    "3f. Foreign-born mother\nnative speaker",
    "4f. Foreign-born no children female\nnative speaker",
    "5f. Native-born mother\nnative speaker",
    "6f. Native-born no children female\nnative speaker"
  ),
  AME = c(AME1, AME2, AME3, AME4, AME5, AME6),
  lower = c(ttest1$conf.int[1], ttest2$conf.int[1], ttest3$conf.int[1],
    ttest4$conf.int[1], ttest5$conf.int[1], ttest6$conf.int[1]),
  upper = c(ttest1$conf.int[2], ttest2$conf.int[2], ttest3$conf.int[2],
    ttest4$conf.int[2], ttest5$conf.int[2], ttest6$conf.int[2])
)

ame_df <- ame_df %>%
  mutate(
    AME = -AME,
    lower = lower,
    upper = upper
  )

ame_dfm <- data.frame(
  type = c(
    "1m. Foreign-born father\nnon-native speaker",
    "2m. Foreign-born no children male\nnon-native speaker",
    "3m. Foreign-born father\nnative speaker",
    "4m. Foreign-born no children male\nnative speaker",
    "5m. Native-born father\nnative speaker",
    "6m. Native-born no children male\nnative speaker"
  ),
  AME = c(AME1m, AME2m, AME3m, AME4m, AME5m, AME6m),

```

```

lower = c(ttest1m$conf.int[1], ttest2m$conf.int[1], ttest3m$conf.int[1],
          ttest4m$conf.int[1], ttest5m$conf.int[1], ttest6m$conf.int[1]),
upper = c(ttest1m$conf.int[2], ttest2m$conf.int[2], ttest3m$conf.int[2],
          ttest4m$conf.int[2], ttest5m$conf.int[2], ttest6m$conf.int[2])
)

ame_dfm <- ame_dfm %>%
  mutate(
    AME = -AME,
    lower = lower,
    upper = upper
  )

ame_all <- bind_rows(
  ame_df %>% mutate(sex = "f"),
  ame_dfm %>% mutate(sex = "m")
)

ame_all <- ame_all %>%
  mutate(type = factor(type, levels = rev(type)))

pal12 <- viridis(n = nlevels(ame_all$type), option = "D", begin = 0.1, end =
0.9)

# extract numeric prefix (1-6) and sex (f/m)
ame_all <- ame_all %>%
  mutate(
    idx = as.integer(substr(type, 1, 1)),
    sex = ifelse(grepl("^\\dm\\.", type), "m", "f")
  )

# build desired order: 1f,1m,2f,2m,...
type_order <- ame_all %>%
  distinct(type, idx, sex) %>%
  arrange(idx, sex) %>%           # f comes before m alphabetically
  pull(type)

# apply order (reverse because coord_flip)
ame_all$type <- factor(ame_all$type, levels = rev(type_order))

```

Fig 1 Old Style orig CIs

```

agg_png(here("Results", "Fig1.png"), res = 144, height = 750, width = 750)

ggplot(ame_all, aes(x = type, y = AME, fill = type)) +
  geom_col(width = 0.55, color = "black") +           # thinner bars
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  scale_fill_manual(values = pal12) +                # 12 viridis colors
  labs(

```

```

x = NULL,
y = "Average Treatment Effect\nof Participation in ALE",
fill = "Type"
) +
coord_flip() +
theme_minimal(base_size = 14) +
theme(
  axis.text.y = element_text(hjust = 1, size = 8),
  axis.text.x = element_text(size = 8),
  axis.title.x = element_text(size = 8),
  legend.position = "none"
)

dev.off()

## png
## 2

include_graphics(here("Results", "Fig1.png"))

```

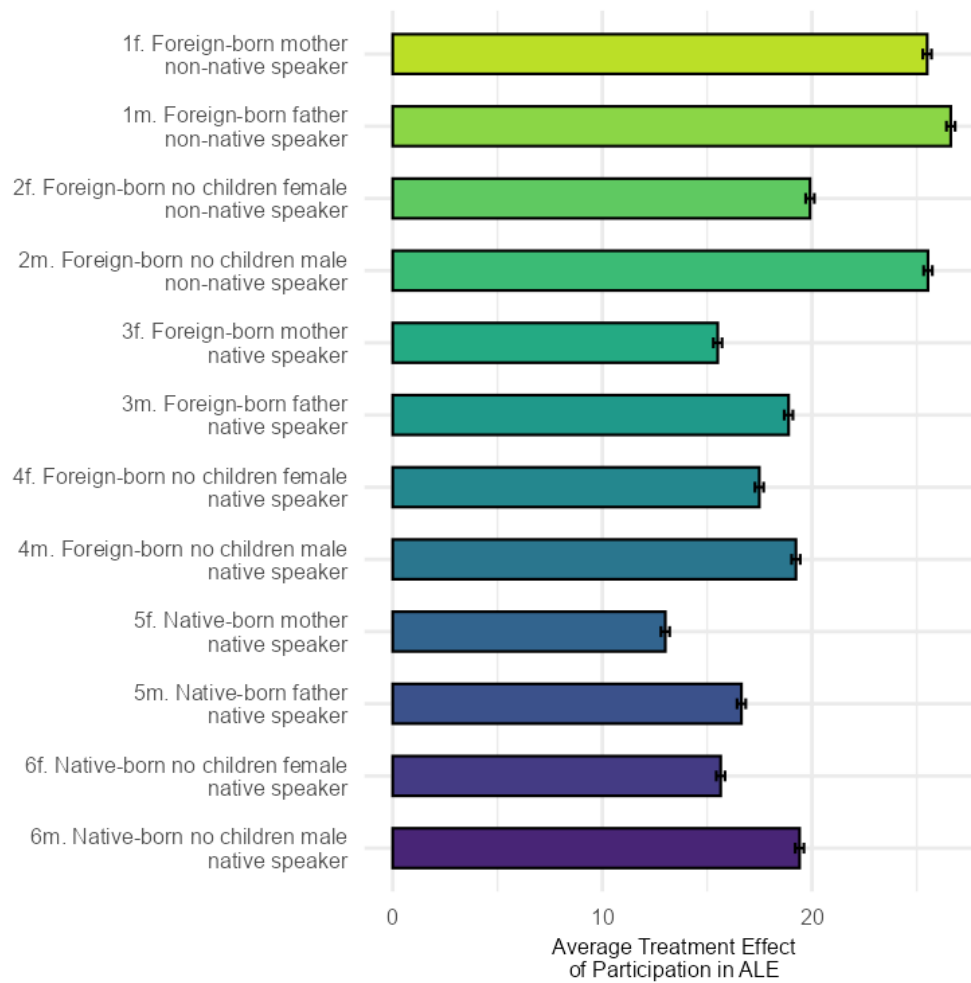


Fig 2. ATEs,

circle labels

```

ame_all <- bind_rows(
  ame_df %>% mutate(sex = "f"),
  ame_dfm %>% mutate(sex = "m")
) %>%
  mutate(
    idx = as.integer(substr(type, 1, 1)),
    sex = ifelse(grepl("^\\dm\\. ", type), "m", "f")
  )

type_order <- ame_all %>%
  distinct(type, idx, sex) %>%
  arrange(idx, sex) %>%
  pull(type)

ame_all <- ame_all %>%
  mutate(
    type = factor(type, levels = rev(type_order)) # reverse for top-to-
bottom display
  )

# Add binary indicators for the dot matrix

ame_all <- ame_all %>%
  mutate(
    Female = ifelse(sex == "f", 1, 0),
    `Foreign-born` = case_when(
      idx %in% c(1, 2, 3, 4) ~ 1,
      idx %in% c(5, 6) ~ 0
    ),
    `Non-native-speaker` = case_when(
      idx %in% c(1, 2) ~ 1,
      idx %in% c(3, 4, 5, 6) ~ 0
    ),
    Parent = case_when(
      idx %in% c(1, 3, 5) ~ 1,
      idx %in% c(2, 4, 6) ~ 0
    )
  )

# Convert for dots

dot_df <- ame_all %>%
  select(type, Female, `Foreign-born`, `Non-native-speaker`, Parent) %>%
  pivot_longer(
    cols = c(Female, `Foreign-born`, `Non-native-speaker`, Parent),
    names_to = "variable",
    values_to = "value"
  ) %>%

```

```

mutate(
  variable = factor(
    variable,
    levels = c("Female", "Foreign-born", "Non-native-speaker", "Parent")
  )
)

# New Labelling with dots in a matrix

p_dots <- ggplot(dot_df, aes(x = variable, y = type)) +
  geom_point(
    data = subset(dot_df, value == 0),
    shape = 21,
    size = 3.2,
    stroke = 0.5,
    fill = "white",
    color = "black"
  ) +
  geom_point(
    data = subset(dot_df, value == 1),
    shape = 16,
    size = 3.2,
    color = "black"
  ) +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.text.x = element_text(size = 10, angle = 45, hjust = 1)
  )

p_bar <- ggplot(ame_all, aes(x = type, y = AME, fill = type)) +
  geom_col(width = 0.55, color = "black") +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  scale_fill_manual(values = pal12) +
  scale_y_continuous(breaks = seq(0, 60, by = 5)) +
  coord_flip() +
  labs(
    x = NULL,
    y = "Average Treatment Effect\nof Participation in ALE"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),

```

```

    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none"
  )

# Combine panels

#p_final <- p_dots + p_bar +
# plot_layout(widths = c(1.1, 2.4))

agg_png(here("Results", "Fig1_dotmatrix.png"), res = 144, height = 750, width
= 1100)

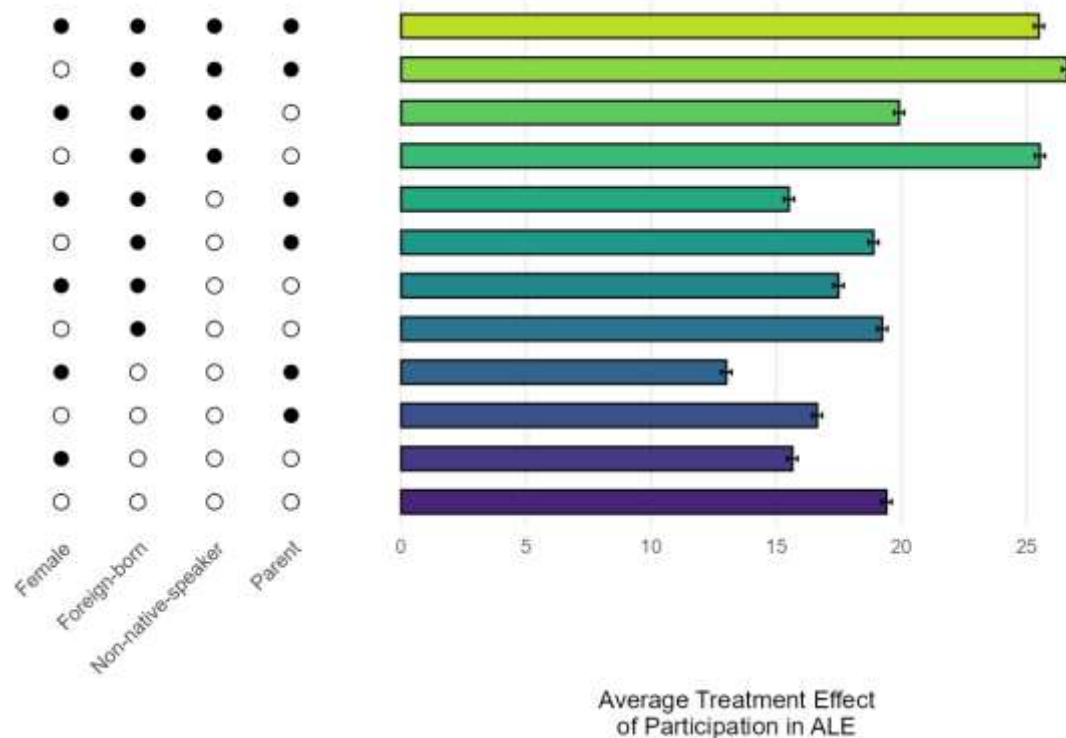
ggarrange(
  p_dots,
  p_bar,
  ncol = 2,
  widths = c(1.1, 2.4),
  align = "h"
)

dev.off()

## png
## 2

include_graphics(here("Results", "Fig1_dotmatrix.png"))

```



Adjust CIs

The confidence intervals here refer to the entire sample across 26 countries. To make them more realistic to a treatment in one country, we adjust them down to a sample of 5,700 which is the average sample size per country.

```
# Function to adjust CI to a new sample size (e.g., average country = 5700)
adjust_ci <- function(ttest_obj, new_n = 5700) {
  mean_diff <- ttest_obj$estimate[1] - ttest_obj$estimate[2]
  pooled_sd <- sqrt(((ttest_obj$parameter + 1) * ttest_obj$stderr^2)) #
  back-calculate SD
  se_new <- pooled_sd / sqrt(new_n)
  ci_margin <- qt(0.975, df = new_n - 1) * se_new
  lower <- mean_diff - ci_margin
  upper <- mean_diff + ci_margin
  return(c(lower, upper))
}

# Apply adjusted CI to each t-test result
ci1 <- adjust_ci(ttest1)
ci2 <- adjust_ci(ttest2)
ci3 <- adjust_ci(ttest3)
ci4 <- adjust_ci(ttest4)
ci5 <- adjust_ci(ttest5)
ci6 <- adjust_ci(ttest6)

ci1m <- adjust_ci(ttest1m)
ci2m <- adjust_ci(ttest2m)
ci3m <- adjust_ci(ttest3m)
ci4m <- adjust_ci(ttest4m)
ci5m <- adjust_ci(ttest5m)
ci6m <- adjust_ci(ttest6m)

# add single country CIs 'sCIs'
ame_all <- ame_all %>%
  mutate(upper_sCI = c(ci1[2], ci2[2], ci3[2], ci4[2], ci5[2], ci6[2],
    ci1m[2], ci2m[2], ci3m[2], ci4m[2], ci5m[2], ci6m[2]),
    lower_sCI = c(ci1[1], ci2[1], ci3[1], ci4[1], ci5[1], ci6[1],
    ci1m[1], ci2m[1], ci3m[1], ci4m[1], ci5m[1], ci6m[1])
  )
```

Fig 1 new CIs

Makes no difference

```
agg_png(here("Results", "Fig1_sCI.png"), res = 144, height = 750, width =
750)

ggplot(ame_all, aes(x = type, y = AME, fill = type)) +
  geom_col(width = 0.55, color = "black") + # thinner bars
```

```

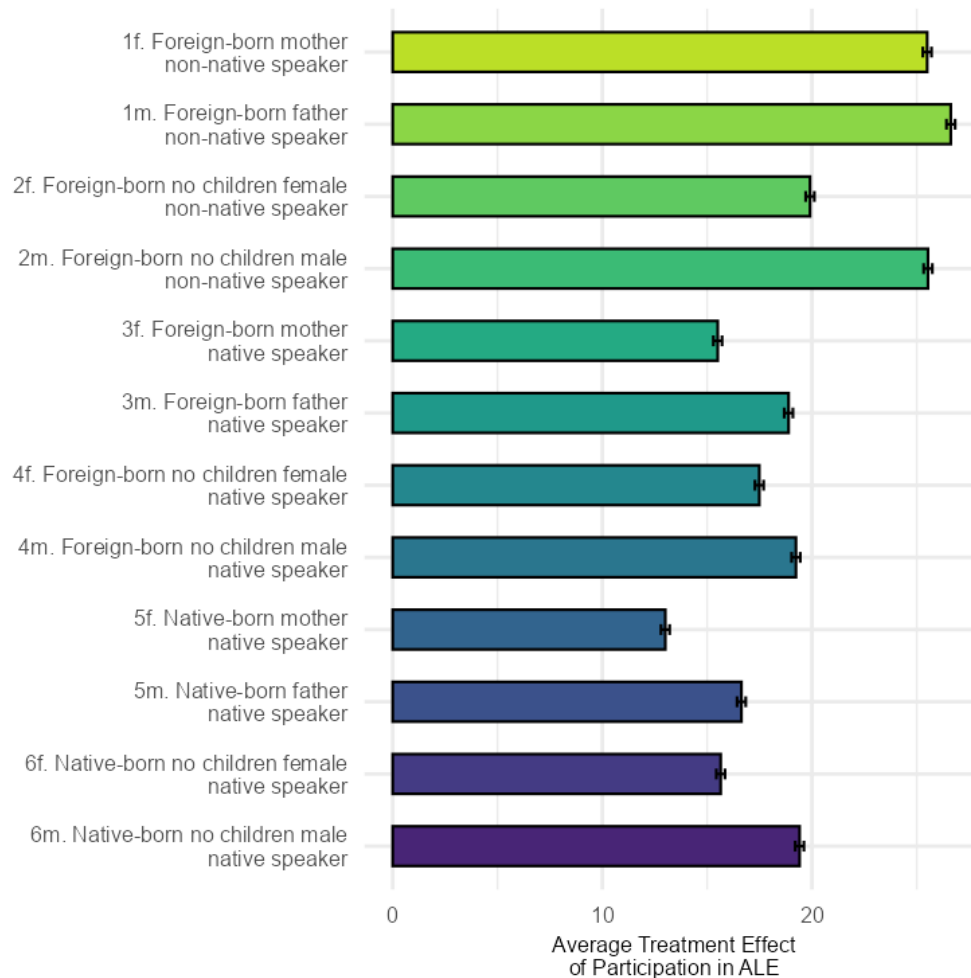
geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
scale_fill_manual(values = pal12) + # 12 viridis colors
labs(
  x = NULL,
  y = "Average Treatment Effect\nof Participation in ALE",
  fill = "Type"
) +
coord_flip() +
theme_minimal(base_size = 14) +
theme(
  axis.text.y = element_text(hjust = 1, size = 8),
  axis.text.x = element_text(size = 8),
  axis.title.x = element_text(size = 8),
  legend.position = "none"
)

dev.off()

## png
## 2

include_graphics(here("Results", "Fig1_sCI.png"))

```



Population Predictions

Predictions

Many ideas supported by interactions with ChatGPT and Google Gemini

```
# Get variable names as vectors
vars_raw    <- c("female", "FORBORN", "FORLANG", "children")
vars_within <- c("female_within", "forborn_within", "forlang_within",
"children_within")
vars_between <- c("female_between", "forborn_between", "forlang_between",
"children_between")

# Map within/between names used in the df
map_within <- c(female="female_within", FORBORN="forborn_within",
FORLANG="forlang_within", children="children_within")
map_between <- c(female="female_between", FORBORN="forborn_between",
FORLANG="forlang_between", children="children_between")

# Build counterfactual newdata for predict(m4) consistent with within/between
coding
```

```

make_newdata_for_m4 <- function(df_base, spec, nfe_value) {
  stopifnot(all(names(spec) %in% vars_raw))
  nd <- df_base

  # Set NFE12 within directly to 0/1 (this is "within", so this is a strong
  intervention).

  nd$NFE12_within <- nfe_value - nd$NFE12_between

  # For each targeting var: if spec is 0/1, set within = target - between;
  else leave as observed (population mix).
  for (v in vars_raw) {
    target <- spec[[v]]
    if (!is.na(target)) {
      w <- map_within[[v]]
      b <- map_between[[v]]
      nd[[w]] <- target - nd[[b]]
    }
  }

  nd
}

# Compute AME for a spec: E[pred(NFE=1) - pred(NFE=0)] (weighted)
calc_ame_spec <- function(model_m4, df_base, spec, weight_var="aweight") {
  nd0 <- make_newdata_for_m4(df_base, spec, nfe_value = 0)
  nd1 <- make_newdata_for_m4(df_base, spec, nfe_value = 1)

  p0 <- predict(model_m4, newdata = nd0, re.form = NA)
  p1 <- predict(model_m4, newdata = nd1, re.form = NA)

  d <- p1 - p0 # gain from ALE

  w <- df_base[[weight_var]]
  ok <- is.finite(d) & is.finite(w) & !is.na(w) & w > 0
  d <- d[ok]; w <- w[ok]

  ame <- weighted.mean(d, w)

  # Simple weighted SE (design-based would be better; this is a pragmatic
  approximation)
  se <- sqrt( sum(w^2 * (d - ame)^2) / (sum(w)^2) )
  ci <- ame + c(-1, 1) * 1.96 * se

  list(AME = ame, SE = se, lower = ci[1], upper = ci[2])
}

# Identify who is "in the targeted group" given a spec (raw vars and then NA
means ignore)

```

```

in_group <- function(df, spec) {
  idx <- rep(TRUE, nrow(df))
  for (v in vars_raw) {
    target <- spec[[v]]
    if (!is.na(target)) idx <- idx & (df[[v]] == target)
  }
  idx
}

# Population gain per country from targeting spec:
# Boost skill_reading by AME for those in group AND NFE12==0, then take
# weighted mean difference.
# Population gain per country from targeting spec
calc_country_gains <- function(df2_full, spec, AME, ctx_data,
weight_var="aweight") {

  df <- df2_full %>%
    left_join(ctx_data, by = "iso3c") %>%
    mutate(context_effect = replace_na(context_effect, 0))

  g <- in_group(df, spec)
  g[is.na(g)] <- FALSE

  treatable <- g & df$NFE12 == 0 & !is.na(df$skill_reading)
  treatable[is.na(treatable)] <- FALSE

  df <- df %>%
    mutate(
      net_effect = pmax(0, AME + context_effect),
      skill_boost = ifelse(treatable, skill_reading + net_effect,
skill_reading),
      g_total = ifelse(g, 1, 0),
      g_noale = ifelse(g & NFE12 == 0, 1, 0)
    )

  df$g_noale[is.na(df$g_noale)] <- 0

  out <- df %>%
    filter(!is.na(iso3c)) %>%
    group_by(iso3c) %>%
    summarise(
      skill_base = weighted.mean(skill_reading, .data[[weight_var]], na.rm =
TRUE),
      skill_gain = weighted.mean(skill_boost, .data[[weight_var]], na.rm =
TRUE) - skill_base,
      pct_group = weighted.mean(g_total, .data[[weight_var]], na.rm =
TRUE),
      pct_group_noale = weighted.mean(g_noale, .data[[weight_var]], na.rm =
TRUE),

```

```

    group_skill_base = weighted.mean(skill_reading[g],
.data[[weight_var]][g], na.rm = TRUE),
    pop18to65 = sum(SPFWT0, na.rm = TRUE),
    context_effect = first(context_effect),
    .groups = "drop"
  )

  out
}

```

Create all specifications

```

make_all_specs <- function(vars = vars_raw) {
  # all subsets except empty
  subsets <- unlist(lapply(1:length(vars), function(k) combn(vars, k,
simplify = FALSE)), recursive = FALSE)

  specs <- map_dfr(subsets, function(active) {
    grid <- expand.grid(rep(list(c(0,1)), length(active)))
    colnames(grid) <- active
    # inactive vars are NA ("all")
    for (v in setdiff(vars, active)) grid[[v]] <- NA_integer_
    grid %>% select(all_of(vars))
  })

  specs %>%
    mutate(
      spec_id = row_number(),
      spec_label = pmap_chr(across(all_of(vars)), function(female, FORBORN,
FORLANG, children) {
        lab1 <- if (is.na(female)) "female=all" else paste0("female=",
female)
        lab2 <- if (is.na(FORBORN)) "FORBORN=all" else paste0("FORBORN=",
FORBORN)
        lab3 <- if (is.na(FORLANG)) "FORLANG=all" else paste0("FORLANG=",
FORLANG)
        lab4 <- if (is.na(children)) "children=all" else paste0("children=",
children)
        paste(lab1, lab2, lab3, lab4, sep="; ")
      })
    )
}

specs_all <- make_all_specs()
nrow(specs_all) # 80

## [1] 80

```

Citations

```
packages %>%
  map(citation) %>%
  print(style = "text")

## [[1]]
## Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R,
## Grolemund G, Hayes A, Henry L, Hester J, Kuhn M, Pedersen TL, Miller E,
## Bache SM, Müller K, Ooms J, Robinson D, Seidel DP, Spinu V, Takahashi
## K, Vaughan D, Wilke C, Woo K, Yutani H (2019). "Welcome to the
## tidyverse." Journal of Open Source Software, 4(43), 1686.
## doi:10.21105/joss.01686 <https://doi.org/10.21105/joss.01686>.
##
## [[2]]
## Wickham H, Hester J, Bryan J (2024). _readr: Read Rectangular Text
## Data_. R package version 2.1.5,
## <https://CRAN.R-project.org/package=readr>.
##
## [[3]]
## Xie Y (2025). _knitr: A General-Purpose Package for Dynamic Report
## Generation in R_. R package version 1.50, <https://yihui.org/knitr/>.
##
## Xie Y (2015). _Dynamic Documents with R and knitr_, 2nd edition.
## Chapman and Hall/CRC, Boca Raton, Florida. ISBN 978-1498716963,
## <https://yihui.org/knitr/>.
##
## Xie Y (2014). "knitr: A Comprehensive Tool for Reproducible Research in
## R." In Stodden V, Leisch F, Peng RD (eds.), _Implementing Reproducible
## Computational Research_. Chapman and Hall/CRC. ISBN 978-1466561595.
##
## [[4]]
## Müller K (2020). _here: A Simpler Way to Find Your Files_. R package
## version 1.0.1, <https://CRAN.R-project.org/package=here>.
##
## [[5]]
## Wickham H (2016). _ggplot2: Elegant Graphics for Data Analysis_.
## Springer-Verlag New York. ISBN 978-3-319-24277-4,
## <https://ggplot2.tidyverse.org>.
##
## [[6]]
## Slowikowski K (2024). _ggrepel: Automatically Position Non-Overlapping
## Text Labels with 'ggplot2'_. R package version 0.9.6,
## <https://CRAN.R-project.org/package=ggrepel>.
##
## [[7]]
## Pedersen T, Shemanarev M (2024). _ragg: Graphic Devices Based on AGG_.
## R package version 1.3.3, <https://CRAN.R-project.org/package=ragg>.
##
## [[8]]
## Zhu H (2024). _kableExtra: Construct Complex Table with 'kable' and
```

```

## Pipe Syntax_. R package version 1.4.0,
## <https://CRAN.R-project.org/package=kableExtra>.
##
## [[9]]
## Arel-Bundock V, Enevoldsen N, Yetman C (2018). "countrycode: An R
## package to convert country names and country codes." Journal of Open
## Source Software_, *3*(28), 848. <https://doi.org/10.21105/joss.00848>.
##
## [[10]]
## Wickham H, Bryan J (2025). _readxl: Read Excel Files_. R package
## version 1.4.5, <https://CRAN.R-project.org/package=readxl>.
##
## [[11]]
## Long JA (2022). _jtools: Analysis and Presentation of Social Scientific
## Data_. R package version 2.2.0,
## <https://cran.r-project.org/package=jtools>.
##
## [[12]]
## Bates D, Mächler M, Bolker B, Walker S (2015). "Fitting Linear
## Mixed-Effects Models Using lme4." Journal of Statistical Software_,
## *67*(1), 1-48. doi:10.18637/jss.v067.i01
## <https://doi.org/10.18637/jss.v067.i01>.
##
## [[13]]
## Kuznetsova A, Brockhoff PB, Christensen RHB (2017). "lmerTest Package:
## Tests in Linear Mixed Effects Models." Journal of Statistical
## Software_, *82*(13), 1-26. doi:10.18637/jss.v082.i13
## <https://doi.org/10.18637/jss.v082.i13>.
##
## [[14]]
## Bolker B, Robinson D (2024). _broom.mixed: Tidying Methods for Mixed
## Models_. R package version 0.2.9.6,
## <https://CRAN.R-project.org/package=broom.mixed>.
##
## [[15]]
## Garnier, Simon, Ross, Noam, Rudis, Robert, Camargo, Pedro A, Sciaini,
## Marco, Scherer, Cédric (2024). _viridis(Lite) - Colorblind-Friendly
## Color Maps for R_. doi:10.5281/zenodo.4679423
## <https://doi.org/10.5281/zenodo.4679423>, viridis package version
## 0.6.5, <https://sjmgarnier.github.io/viridis/>.
##
## [[16]]
## Lumley T (2024). "survey: analysis of complex survey samples." R
## package version 4.4.
##
## Lumley T (2004). "Analysis of Complex Survey Samples." Journal of
## Statistical Software_, *9*(1), 1-19. R package version 2.2.
##
## Lumley T (2010). _Complex Surveys: A Guide to Analysis Using R: A Guide
## to Analysis Using R_. John Wiley and Sons.

```

```
##
## [[17]]
## Kassambara A (2023). _ggpubr: 'ggplot2' Based Publication Ready Plots_.
## R package version 0.6.0, <https://CRAN.R-project.org/package=ggpubr>.
##
## [[18]]
## Wickham H, Henry L (2025). _purrr: Functional Programming Tools_. R
## package version 1.0.4, <https://CRAN.R-project.org/package=purrr>.
```

Colophon

sessionInfo()

```
## R version 4.4.1 (2024-06-14 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
##  [1] ggpubr_0.6.0      survey_4.4-2      survival_3.6-4
##  [4] viridis_0.6.5     viridisLite_0.4.2 broom.mixed_0.2.9.6
##  [7] lmerTest_3.1-3    lme4_1.1-37      Matrix_1.7-0
## [10] jtools_2.3.0      readxl_1.4.5     countrycode_1.6.0
## [13] kableExtra_1.4.0  ragg_1.3.3       ggrepel_0.9.6
## [16] here_1.0.1        knitr_1.50       lubridate_1.9.4
## [19] forcats_1.0.0     stringr_1.5.1    dplyr_1.1.4
## [22] purrr_1.0.4       readr_2.1.5      tidyr_1.3.1
## [25] tibble_3.2.1      ggplot2_3.5.1    tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.3      DBI_1.2.3         gridExtra_2.3
##  [4] rlang_1.1.5       magrittr_2.0.3    furrr_0.3.1
##  [7] compiler_4.4.1    mgcv_1.9-1        png_0.1-8
## [10] systemfonts_1.2.1 vctrs_0.6.5       pkgconfig_2.0.3
## [13] crayon_1.5.3      fastmap_1.2.0     backports_1.5.0
```

## [16]	labeling_0.4.3	pander_0.6.6	rmarkdown_2.29
## [19]	tzdb_0.5.0	nloptr_2.2.1	xfun_0.51
## [22]	uuid_1.2-1	broom_1.0.7	parallel_4.4.1
## [25]	R6_2.6.1	stringi_1.8.4	parallelly_1.45.0
## [28]	car_3.1-3	boot_1.3-30	cellranger_1.1.0
## [31]	numDeriv_2016.8-1.1	Rcpp_1.0.14	assertthat_0.2.1
## [34]	pacman_0.5.1	splines_4.4.1	timechange_0.3.0
## [37]	tidyselect_1.2.1	rstudioapi_0.17.1	abind_1.4-8
## [40]	yaml_2.3.10	codetools_0.2-20	listenv_0.9.1
## [43]	lattice_0.22-6	withr_3.0.2	flextable_0.9.7
## [46]	askpass_1.2.1	evaluate_1.0.3	future_1.58.0
## [49]	huxtable_5.6.0	zip_2.3.2	xml2_1.3.8
## [52]	pillar_1.10.1	carData_3.0-5	reformulas_0.4.0
## [55]	generics_0.1.3	rprojroot_2.0.4	hms_1.1.3
## [58]	munsell_0.5.1	scales_1.3.0	minqa_1.2.8
## [61]	globals_0.18.0	glue_1.8.0	gdtools_0.4.2
## [64]	tools_4.4.1	data.table_1.17.0	ggsignif_0.6.4
## [67]	cowplot_1.1.3	mitools_2.4	rbibutils_2.3
## [70]	colorspace_2.1-1	nlme_3.1-164	Formula_1.2-5
## [73]	cli_3.6.4	textshaping_1.0.0	officer_0.6.8
## [76]	fontBitstreamVera_0.1.1	svglite_2.1.3	gtable_0.3.6
## [79]	rstatix_0.7.2	fontquiver_0.2.1	digest_0.6.37
## [82]	farver_2.1.2	htmltools_0.5.8.1	lifecycle_1.0.4
## [85]	fontLiberation_0.1.0	openssl_2.3.2	MASS_7.3-60.2